

Finite-range overcomplete Wannier measurements and bulk topology

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Overcomplete Wannier modes provide a local description of the occupations of a Chern band, but their exact spatial tails cannot be retained in a finite-range circuit. We explain what changes when these modes are truncated. Starting from the exact occupied projector, we derive the effective form factors and the auxiliary Hamiltonian built from the truncated modes. We then give two sufficient conditions under which band mixing preserves the reference Chern number and evaluate the shortest nontrivial square window in the model used for adaptive preparation. The onsite limit provides a useful contrast: an overlap zero can survive even though the auxiliary band is trivial. Finally, we distinguish these reference-state results from the topology of states produced by conditioned measurements, including trajectories whose particle number changes.

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I. THE IDEAL CONSTRUCTION

The adaptive circuit fills modes associated with the lower band and empties modes associated with the upper band. With exact overcomplete Wannier (OW) functions, these conditions select the intended Chern insulator. With truncated functions, the two sets of conditions are no longer generally compatible. The question is whether losing exact stabilization also requires losing the bulk topology. To address this question, we first identify the state that the untruncated construction is designed to prepare, following the projection construction used in BPJ [1].

We consider two orbitals per unit cell and set the lattice spacing to one unless it is displayed explicitly. The single-particle Hamiltonian and band projectors are

$$h(\mathbf{k}) = \mathbf{n}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad \mathbf{n}(\mathbf{k}) = (\sin k_x, \sin k_y, \alpha - \cos k_x - \cos k_y), \quad P_{\pm}(\mathbf{k}) = \frac{\mathbf{1} \pm \hat{\mathbf{n}}(\mathbf{k}) \cdot \boldsymbol{\sigma}}{2}. \quad (1)$$

Here $\hat{\mathbf{n}} = \mathbf{n}/|\mathbf{n}|$ is a unit vector; the hat does not denote an operator. Our orientation convention assigns $\mathfrak{C}[P_-] = 1$ for $0 < \alpha < 2$. The many-body state $|\text{CI}\rangle$ occupies the lower band and has one fermion per unit cell.

Choose the two orthonormal trial orbitals $\tau_A = (1, 1)^\top/\sqrt{2}$ and $\tau_B = (1, -1)^\top/\sqrt{2}$. In the full single-particle Hilbert space, write the normalized OW states as

$$|W_{\mathbf{r},\nu,\pm}\rangle = \frac{P_{\pm}|\mathbf{r},\tau_{\nu}\rangle}{\sqrt{Z_{\nu,\pm}}}, \quad Z_{\nu,\pm} = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \tau_{\nu}^{\dagger} P_{\pm}(\mathbf{k}) \tau_{\nu} = \frac{1}{2}. \quad (2)$$

The last equality follows from reflection symmetry: the only Pauli component seen by either trial orbital is σ^x , and $\int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \hat{n}_x = 0$. The physical orbital index is μ ; $\nu = A, B$ labels the two OW families, not two new physical orbitals.

Completeness of the trial orbitals gives a particularly simple reconstruction:

$$\frac{1}{2} \sum_{\mathbf{r},\nu} |W_{\mathbf{r},\nu,-}\rangle \langle W_{\mathbf{r},\nu,-}| = P_- \left(\sum_{\mathbf{r},\nu} |\mathbf{r},\tau_{\nu}\rangle \langle \mathbf{r},\tau_{\nu}| \right) P_- = P_-^2 = P_-. \quad (3)$$

Thus the lower-band OW modes form a tight frame. The factor $1/2$ compensates for their redundancy; summing their projectors without this factor gives $2P_-$, not a physical occupation matrix.

To see the role of the two families, choose a normalized Bloch eigenvector $|\psi_-(\mathbf{k})\rangle$ on a local momentum patch. Its OW form factors are

$$f_{\nu,-}(\mathbf{k}) = \sqrt{2} \langle \psi_-(\mathbf{k}) | \tau_{\nu} \rangle, \quad \sum_{\nu=A,B} |f_{\nu,-}(\mathbf{k})|^2 = 2. \quad (4)$$

An individual form factor has zeros, but the two cannot vanish at the same momentum. One family supplies the band component missed by the other. The projector is globally smooth even though a normalized occupied eigenvector cannot be chosen smooth and periodic everywhere. Projecting local trial orbitals therefore produces exponentially localized, nonorthogonal modes without constructing a forbidden orthonormal Wannier basis [2].

Let $i = (\mathbf{r}, \mu)$ and $\{\hat{c}_i, \hat{c}_j^{\dagger}\} = \delta_{ij}$. The mode operator is $\hat{\chi}_j = \sum_i W_j(i)^* \hat{c}_i$, where j collects the center, family, and band labels. Its number operator $\hat{N}_j = \hat{\chi}_j^{\dagger} \hat{\chi}_j$ is a projector. The target satisfies

$$\hat{N}_{\mathbf{r},\nu,-} |\text{CI}\rangle = |\text{CI}\rangle, \quad (\hat{1} - \hat{N}_{\mathbf{r},\nu,+}) |\text{CI}\rangle = |\text{CI}\rangle. \quad (5)$$

Distinct OW number operators need not commute. Nevertheless, these exact conditions are compatible because the two families span the occupied band, while all upper-band modes are orthogonal to it.

We will distinguish the modes defining the operations from the state on which they act. In BPJ notation the latter has correlation matrix

$$(G_{\mathbf{m}})_{ij} = \text{Tr}(\hat{\rho}_{\mathbf{m}} \hat{c}_i^{\dagger} \hat{c}_j), \quad \Gamma_{\mathbf{m}} = G_{\mathbf{m}}^{\top}, \quad \Gamma_{\mathbf{m}}^2 = \Gamma_{\mathbf{m}} \quad (6)$$

for a pure number-conserving Gaussian trajectory with measurement record $\mathbf{m}$. The transpose converts to the single-particle projector ordering. For the ideal target $\Gamma = P_-$. A typical conditioned state need not be translation invariant, so it need not have a rank-one occupied band at each momentum. Nor is the record average $\bar{\Gamma}$ generally a projector.

II. WHAT A FINITE WINDOW CHANGES

We now truncate the OW wavefunction rather than its outer product. Define

$$P_-(\mathbf{d}) = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} e^{i\mathbf{k}\cdot\mathbf{d}} P_-(\mathbf{k}), \quad g_w(\mathbf{d}) = \begin{cases} 1, & |d_x|, |d_y| \leq w, \\ 0, & \text{otherwise,} \end{cases} \quad w = n_{\text{shell}}. \quad (7)$$

With this convention $P_-(\mathbf{k}) = \sum_{\mathbf{d}} e^{-i\mathbf{k}\cdot\mathbf{d}} P_-(\mathbf{d})$. An OW mode centered at $\mathbf{r}$ has coefficients proportional to $P_-(\mathbf{r}' - \mathbf{r}) \tau_{\nu}$. Its truncated, normalized Fourier amplitude is

$$W_{\nu,-;w}(\mathbf{k}) = \frac{F_w(\mathbf{k}) \tau_{\nu}}{\sqrt{Z_{\nu,-;w}}}, \quad F_w(\mathbf{k}) = \sum_{\mathbf{d}} g_w(\mathbf{d}) P_-(\mathbf{d}) e^{-i\mathbf{k}\cdot\mathbf{d}}, \quad Z_{\nu,-;w} = \sum_{\mathbf{d}} g_w(\mathbf{d})^2 \|P_-(\mathbf{d}) \tau_{\nu}\|^2. \quad (8)$$

The normalization is in real space, equivalently $\int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \|W_{\nu,-;w}(\mathbf{k})\|^2 = 1$. It is not a normalization separately at each momentum. At finite w , F_w need not be a projector. The amplitude of a translated mode acquires only the phase $e^{-i\mathbf{k}\cdot\mathbf{r}}$; we set its center to zero below.

A. The effective overlap and opposite-band weight

Substituting the Fourier representation of $P_-(\mathbf{d})$ into Eq. (8) gives

$$F_w(\mathbf{k}) = \int_{\text{BZ}} \frac{d^2 q}{(2\pi)^2} \tilde{g}_w(\mathbf{k} - \mathbf{q}) P_-(\mathbf{q}), \quad \tilde{g}_w(\mathbf{q}) = D_w(q_x) D_w(q_y), \quad D_w(q) = \sum_{d=-w}^w e^{-iqd}. \quad (9)$$

Projecting the truncated mode back onto the target lower band then yields

$$f_{\nu,-}^{(w)}(\mathbf{k}) = \langle \psi_-(\mathbf{k}) | W_{\nu,-;w}(\mathbf{k}) \rangle = \frac{1}{\sqrt{Z_{\nu,-;w}}} \int_{\text{BZ}} \frac{d^2 q}{(2\pi)^2} \tilde{g}_w(\mathbf{k} - \mathbf{q}) \langle \psi_-(\mathbf{k}) | \psi_-(\mathbf{q}) \rangle \langle \psi_-(\mathbf{q}) | \tau_\nu \rangle. \quad (10)$$

This is a convolution dressed by the overlap of Bloch states at different momenta. The product inside the integral is well defined even when several momentum patches are required. A change of gauge at $\mathbf{q}$ cancels between its two factors; a change of gauge at $\mathbf{k}$ changes only the phase of the resulting overlap. On a finite periodic lattice, each normalized momentum integral is replaced by $N_{\text{uc}}^{-1} \sum_{\mathbf{k}}$.

There is also an upper-band component. If $b_\nu^{(w)} = \langle \psi_+ | W_{\nu,-;w} \rangle$, completeness gives

$$\int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} |f_{\nu,-}^{(w)}|^2 + \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} |b_\nu^{(w)}|^2 = 1. \quad (11)$$

Thus truncation changes both the in-band form factor and the band content of the measurement mode. The first term need no longer integrate to one. In particular, renormalizing the effective overlap by itself would conceal part of the opposite-band weight. Figure 1 keeps the real-space normalization of Eq. (8).

B. What an overlap zero tells us

The phase of $f_{\nu,-}^{(w)}$ must be studied in a smooth local gauge. On a disk D containing an isolated zero and no gauge singularity, define its boundary winding by

$$\nu_D[f] = \frac{1}{2\pi} \oint_{\partial D} d \arg f. \quad (12)$$

Suppose the untruncated and truncated overlaps are compared in the same local gauge and $\max_{\partial D} |f^{(w)} - f| < \min_{\partial D} |f|$. Then $(1-s)f + sf^{(w)}$ never vanishes on ∂D . Its winding is an integer continuous in s , and hence is unchanged. This preserves the total signed index of the enclosed zeros, not necessarily their number or position. No assumption of complex analyticity in $k_x + ik_y$ is needed.

The local statement should not be confused with a globally periodic scalar winding argument. Under a Bloch gauge change, $f^{(w)}$ transforms with the inverse phase of the target eigenvector. Equivalently, $P_- W_{\nu,-;w}$ is a section of the target band. If it were nonzero everywhere, its normalization would trivialize that band. A nontrivial target therefore still requires a zero somewhere, provided this projected section is not identically zero. Truncation can displace the original zero without removing the target-band obstruction.

By contrast, if the full two-component field $W_{\nu,-;w}(\mathbf{k})$ is nowhere zero, then $W_{\nu,-;w}/\|W_{\nu,-;w}\|$ is a global smooth periodic section of *its own* line bundle, whose Chern number is zero. The overlap zero is then a statement about its relation to the target, not a nonzero Chern number carried by this single compact-support mode. We will see this distinction explicitly already for $w = 0$.

C. Spatial extent and discarded weight

Two elementary measures of the exact normalized mode are

$$\xi_{\text{rms}}^2 = \sum_{\mathbf{d}} |\mathbf{d}|^2 \|W_{\nu,-}(\mathbf{d})\|^2, \quad p_w = \sum_{\mathbf{d}} g_w(\mathbf{d}) \|W_{\nu,-}(\mathbf{d})\|^2 = 2Z_{\nu,-;w}. \quad (13)$$

Here the RMS radius is measured about the nominal trial-orbital cell, not about the probability centroid. At $\alpha = 1$ we find $\xi_{\text{rms}} \simeq 0.673521a$, while $p_1 \simeq 0.992155$ and $p_2 \simeq 0.999137$. The A -family centroid is $\langle \mathbf{r} \rangle \simeq (0, -0.185336)a$, giving

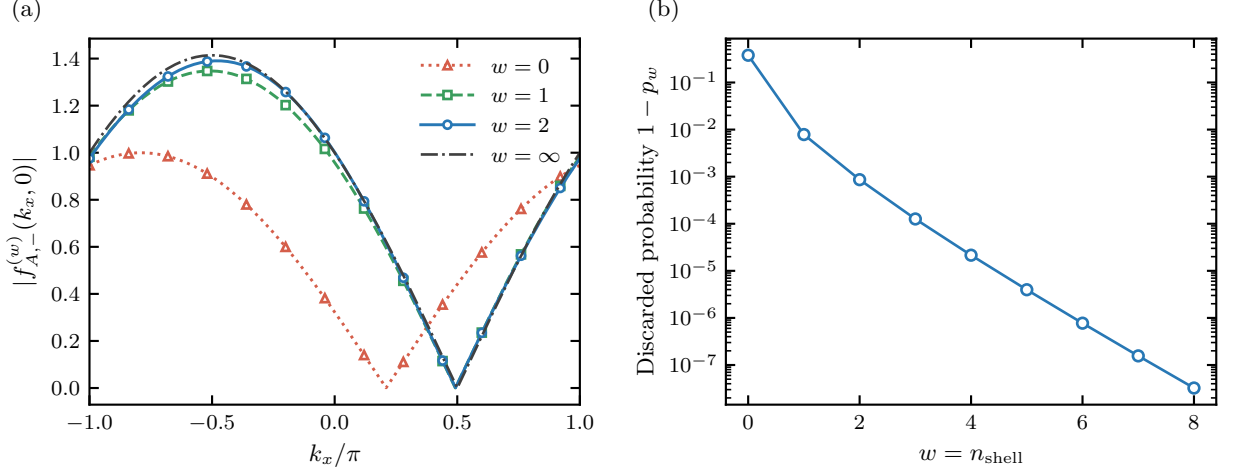


FIG. 1. **Effect of truncating an OW mode.** (a) Lower-band overlap of the A -family mode along $k_y = 0$ at $\alpha = 1$, with unit real-space norm for each mode. (b) Probability discarded by the square window. The form-factor zero moves under truncation even when most weight is retained. These are deterministic band calculations, not trajectory averages.

a centered RMS spread of $0.647520a$. The axial asymptotic tail is controlled instead by the nearest complex-momentum singularity. Here $|\mathbf{n}|^2 = 3 - 2\cos k_x - 2\cos k_y + 2\cos k_x \cos k_y$. Putting $k_x = i\kappa$ and $k_y = \pi$ gives $5 - 4\cosh \kappa = 0$, hence the axial decay length $\xi_{\text{exp},x} = a/\log 2 \simeq 1.442695a$. This tail scale is not the RMS radius and is not an isotropic bound on every direction. A long asymptotic tail can carry very little total weight.

The spread of a band-projected state also depends on quantum geometry. For a normalized amplitude $a(\mathbf{k})$ multiplying a local Bloch state, direct differentiation and Parseval's identity give

$$\langle \mathbf{r}^2 \rangle = \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} [|(-i\nabla_{\mathbf{k}} - \mathcal{A})a|^2 + \text{tr } g(\mathbf{k})|a|^2], \quad \mathcal{A}_\mu = i\langle \psi_- | \partial_\mu \psi_- \rangle, \quad g_{\mu\nu} = \text{Re} \langle \partial_\mu \psi_- | (1 - P_-) | \partial_\nu \psi_- \rangle. \quad (14)$$

The patch transformations of a and $\mathcal{A}$ make this expression gauge invariant. Minimizing it is a geometry-dependent spectral problem for $(-i\nabla - \mathcal{A})^2 + \text{tr } g$, acting on amplitudes with the appropriate patch transitions, as developed by Li *et al.* [2]. The geometric data cannot in general be replaced by the Chern number alone. In particular, a spread bound derived for homogeneous ideal geometry is not a universal minimum truncation radius for our lattice model. We use the directly computed p_w and projector errors instead.

III. THE AUXILIARY HAMILTONIAN OF THE TRUNCATED MODES

To compare an occupied band with the target, we must combine the measurement modes before taking a spectral projection. The relevant translation-summed frame operators are

$$S_{\pm,w}(\mathbf{k}) = \sum_{\nu=A,B} W_{\nu,\pm;w}(\mathbf{k}) W_{\nu,\pm;w}^\dagger(\mathbf{k}). \quad (15)$$

For clarity, a single centered mode $W(\mathbf{d})$ has an outer product with two independent Fourier sums:

$$W(\mathbf{k}) W^\dagger(\mathbf{k}') = \sum_{\mathbf{d}, \mathbf{d}'} e^{-i\mathbf{k} \cdot \mathbf{d}} e^{i\mathbf{k}' \cdot \mathbf{d}'} W(\mathbf{d}) W^\dagger(\mathbf{d}'). \quad (16)$$

With normalized finite-volume momentum kets, its matrix element has an additional factor N_{uc}^{-1} . Summing over all translated centers produces $N_{\text{uc}} \delta_{\mathbf{k}, \mathbf{k}'}$, yielding Eq. (15). For one normalized untruncated family W , the corresponding real-space kernel is an autocorrelation,

$$S_w(\mathbf{d}) = \frac{1}{p_w} \sum_{\mathbf{x}} g_w(\mathbf{x} + \mathbf{d}) g_w(\mathbf{x}) W(\mathbf{x} + \mathbf{d}) W^\dagger(\mathbf{x}), \quad (17)$$

where p_w is its retained probability. Both window factors are necessary. For one wavefunction family its range can reach $2w$; it is not obtained by simply multiplying the original frame kernel by $g_w(\mathbf{d})$.

Since $g_w(0) = 1$, the truncated upper-band amplitude is $(\mathbf{1} - F_w)\tau_\nu/\sqrt{Z_{\nu,+;w}}$. For a real inversion-symmetric window, $F_w^\dagger = F_w$. Keeping potentially unequal normalizations gives

$$S_{-,w} = F_w D_{-,w} F_w^\dagger, \quad S_{+,w} = (\mathbf{1} - F_w) D_{+,w} (\mathbf{1} - F_w)^\dagger, \quad D_{\pm,w} = \sum_\nu \frac{\tau_\nu \tau_\nu^\dagger}{Z_{\nu,\pm;w}}. \quad (18)$$

This is the appropriate expression if the window or trial orbitals break the normalization symmetry.

For the square window in Eq. (7), write $F_w = (\mathbf{1} - \mathbf{d}_w \cdot \boldsymbol{\sigma})/2$. The trace is one because the trace of $P_-(\mathbf{d})$ is $\delta_{\mathbf{d},0}$. The four norms are equal: reflection makes $\int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} d_{w,x} = 0$, while $\tau_{A,B}$ have zero $\sigma^{y,z}$ expectation values. Consequently,

$$Z_{\nu,\pm;w} = Z_w = \frac{1 + \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} |\mathbf{d}_w|^2}{4}, \quad S_{-,w} = \frac{F_w^2}{Z_w}, \quad S_{+,w} = \frac{(\mathbf{1} - F_w)^2}{Z_w}. \quad (19)$$

We define the signed auxiliary Hamiltonian and its occupied projector by

$$h_w = S_{+,w} - S_{-,w} = \frac{\mathbf{1} - 2F_w}{Z_w}, \quad R_w = \mathbf{1}_{(-\infty,0)}(h_w) = \mathbf{1}_{(1/2,\infty)}(F_w), \quad (20)$$

whenever the cutoff is gapped. The simplification of h_w follows from cancellation of the quadratic terms; it does not replace the two-index Fourier transform by a one-index transform. It is special to the equal-norm construction. In the untruncated limit $Z_\infty = 1/2$, $h_\infty = 2(\mathbf{1} - 2P_-)$, and $R_\infty = P_-$.

This construction also explains why a finite-range auxiliary Hamiltonian can have a nontrivial band. Its spectral projection R_w is nonlinear in F_w and generally has exponential spatial tails, even though h_w has finite range. It is not a compactly supported spanning set for the occupied band. The compact-support obstruction of Dubail and Read therefore does not forbid such a Hamiltonian: redundancy alone cannot provide compact modes confined to and spanning an exact nontrivial band, but our truncated modes leak out of that band [3]. The objects F_w , $S_{\pm,w}$, and R_w must not be identified with one another.

IV. WHY BAND MIXING CAN PRESERVE TOPOLOGY

For a smooth periodic occupied projector $P(\mathbf{k})$, we use

$$\mathfrak{C}[P] = \frac{1}{2\pi i} \int_{\text{BZ}} d^2 k \operatorname{tr}(P[\partial_{k_x} P, \partial_{k_y} P]). \quad (21)$$

Its integer character can be seen by choosing local occupied eigenvectors. On an overlap they differ by a phase $e^{i\theta}$, so the connection one-forms $\langle \psi | d\psi \rangle$ differ by $i d\theta$. Stokes' theorem reduces the integrated curvature to integer transition-function windings. The same expression is continuous under a smooth fixed-rank deformation. An integer cannot vary continuously, so the Chern number is constant along such a path [4]. We next give two ways to construct the required path; neither assumes a global eigenvector.

A. A nonzero overlap is sufficient

Let P and R be smooth periodic rank-one projectors. Locally, write an occupied vector of R as $|v\rangle = a|\psi_-\rangle + b|\psi_+\rangle$, with $|a|^2 + |b|^2 = 1$, and set $P = P_-$. The target overlap and opposite-band weight are

$$\operatorname{tr}(PR) = |a|^2 = 1 - \eta, \quad \eta = \operatorname{tr}[(\mathbf{1} - P)R] = |b|^2. \quad (22)$$

If $\operatorname{tr}(PR) > 0$ at every momentum, the upper-band component can be removed continuously:

$$B_s = P + (1 - s)(\mathbf{1} - P), \quad R_s = \frac{B_s R B_s}{\operatorname{tr}(B_s R B_s)}, \quad \operatorname{tr}(B_s R B_s) = 1 - \eta + (1 - s)^2 \eta > 0. \quad (23)$$

The numerator has rank one and is positive, so division by its trace makes it an orthogonal projector. Moreover, $R_0 = R$ and $PRP = \operatorname{tr}(PR)P$ implies $R_1 = P$. Compactness of the Brillouin torus makes the positive overlap uniformly bounded away from zero. Thus

$$\operatorname{tr}[P(\mathbf{k})R(\mathbf{k})] > 0 \text{ for every } \mathbf{k} \implies \mathfrak{C}[R] = \mathfrak{C}[P]. \quad (24)$$

The physical point is that a coherent admixture of the other band need not change the topology of the occupied line. Different Chern numbers require an orthogonality point, although an orthogonality point alone does not imply different topology. An average bound on η would not suffice: it can miss a small region where $|a|$ vanishes. For several occupied bands, the overlap map must instead have full rank everywhere.

A related distinction appears after coherent quenches: a state's Chern number can remain unchanged even though it is not the ground state of the post-quench Hamiltonian [5]. The nonequilibrium Hall response is another quantity and need not be quantized [6].

B. A controlled truncation keeps the auxiliary gap open

For our auxiliary band, a convenient sufficient condition is

$$\delta_w = \sup_{\mathbf{k}} \|F_w(\mathbf{k}) - P_-(\mathbf{k})\|_{\text{op}} < \frac{1}{2}. \quad (25)$$

Consider $F_{w,s} = P_- + s(F_w - P_-)$ and the unnormalized Hamiltonian $H_{w,s} = \mathbf{1} - 2F_{w,s}$. Each eigenvalue of $F_{w,s}$ lies within $s\delta_w$ of either zero or one. The two intervals do not reach $1/2$, and hence

$$\min_{\mathbf{k},n} |E_n[H_{w,s}(\mathbf{k})]| \geq 1 - 2s\delta_w > 0. \quad (26)$$

The spectral projector above $1/2$ supplies the required smooth fixed-rank path from P_- to R_w . At the endpoint $h_w = H_{w,1}/Z_w$ differs only by a positive scale. We obtain

$$\delta_w < \frac{1}{2} \implies \mathfrak{C}[R_w] = 1, \quad \min_{\mathbf{k},n} |E_n[h_w]| \geq \frac{1 - 2\delta_w}{Z_w}. \quad (27)$$

For the traceless two-band Hamiltonian, the full separation between the bands is twice this minimum absolute energy. At a fixed gapped α , the projector kernel decays exponentially. For the indicator window,

$$\delta_w \leq \sum_{\mathbf{d} \notin [-w,w]^2} \|P_-(\mathbf{d})\|_{\text{op}} \longrightarrow 0 \quad (w \longrightarrow \infty). \quad (28)$$

This bound is independent of the volume and proves stability for sufficiently large finite windows. It does not by itself certify $w = 1$, nor a range uniform through a target gap closing. If the normalization symmetry is absent, one can apply the same spectral argument directly to $h_w - h_\infty$, bounding its uniform operator norm by the distance of the untruncated spectrum from zero. Appendix A gives an alternative estimate from the normalized mode errors.

The retained probability p_w is a useful local diagnostic but is weaker than Eq. (25). It integrates information over momenta, whereas the gap must remain open everywhere. Likewise, two projectors can be close on average without satisfying the pointwise overlap criterion. These distinctions matter when interpreting a very small measured band leakage.

V. RANGE ONE AND THE ONSITE COUNTEREXAMPLE

We evaluate the square-window construction at $\alpha = 1$. The Fourier coefficients are computed on 1024^2 and 2048^2 grids; matrix extrema are then sampled on independent 201^2 and 401^2 momentum grids. Table I reports the highest-resolution values. The zero is refined on $k_y = 0$ in a smooth local frame, rather than identified with the smallest sampled value on a momentum grid. Loops of several radii and resolutions give winding -1 around each listed zero.

For $w = 1$, more than 99% of the mode's probability is retained, the sampled error is approximately 0.09875, and the minimum sampled target overlap exceeds 0.9901. Independently summing the four normalized mode outer products reproduces Eq. (20) to numerical precision. Figure 2 displays the interpolating spectral separation and the small but nonuniform mismatch of its occupied band with the target. These tests give substantial numerical evidence that the shortest nontrivial square window lies well within the stable reference phase. Fourier quadrature and finite momentum sampling do not, however, provide an interval-certified bound on the continuum supremum.

The onsite limit is qualitatively different. Here $w = 0$ retains one unit cell with its two physical orbitals. Reflection symmetry gives

$$F_0 = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} P_-(\mathbf{k}) = \frac{\mathbf{1} - d_0\sigma^z}{2}, \quad d_0 = \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} \hat{n}_z \simeq 0.491169, \quad Z_0 = \frac{1 + d_0^2}{4}. \quad (29)$$

TABLE I. **Square-window checks at $\alpha = 1$.** The effective overlap uses the normalized A -family mode. k_* lies on $k_y = 0$. The last three columns are sampled extrema: they are not rigorous continuum bounds. $E_{\min}$ is the minimum absolute energy of the normalized h_w , not the full band separation.

w	p_w	k_*/π	δ_w^{grid}	$\min \text{tr}(P_- R_w)$	$E_{\min}^{\text{grid}}$
0	0.620624	0.209346	0.745585	0.000000	1.582825
1	0.992155	0.492186	0.098754	0.990190	1.802426
2	0.999137	0.492488	0.033797	0.999004	1.902788

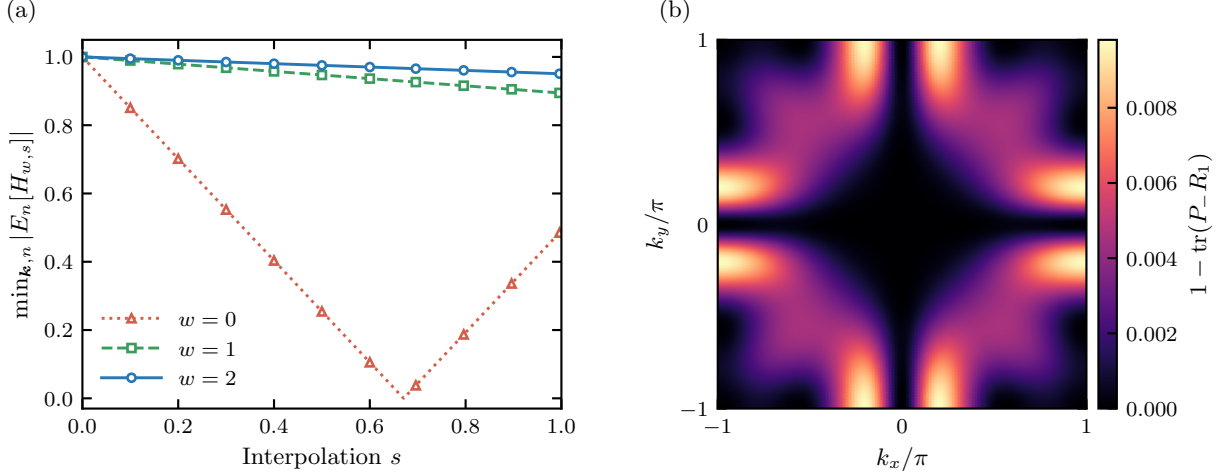


FIG. 2. **Auxiliary-band stability.** (a) Minimum absolute energy of $H_{w,s} = \mathbf{1} - 2[(1-s)P_- + sF_w]$ along the interpolation. The onsite path closes at $s_* = 1/(1+d_0)$; the $w=1,2$ paths remain gapped on the sampled momentum grid. (b) Target-band mismatch $1 - \text{tr}(P_- R_1)$ at $\alpha = 1$. The positive normalization $1/Z_w$ is omitted in (a). These quantities characterize auxiliary bands, not stochastic trajectories.

Therefore $h_0 = (d_0/Z_0)\sigma^z$ and $R_0 = |\downarrow\rangle\langle\downarrow|$ is constant, so $\mathfrak{C}[R_0] = 0$. The onsite auxiliary Hamiltonian is gapped but trivial. At $\mathbf{k} = 0$, the target projector is instead $P_-(0) = |\uparrow\rangle\langle\uparrow|$. Its overlap with R_0 vanishes, and the interpolation has

$$F_{0,s}(0) = \begin{pmatrix} 1 - s(1+d_0)/2 & 0 \\ 0 & s(1+d_0)/2 \end{pmatrix}, \quad s_* = \frac{1}{1+d_0}, \quad H_{0,s_*}(0) = 0. \quad (30)$$

The failure is not just that a sufficient estimate becomes inconclusive: this particular path actually closes its gap.

Nevertheless, the mode $F_0\tau_A/\sqrt{Z_0}$ still has an overlap zero with the target. On $k_y = 0$, a smooth target eigenvector gives

$$f_{A,-}^{(0)}(k_x, 0) = \frac{(1-d_0)\cos(k_x/2) - (1+d_0)\sin(k_x/2)}{2\sqrt{2Z_0}}, \quad \tan \frac{k_*}{2} = \frac{1-d_0}{1+d_0}. \quad (31)$$

Thus $k_*/\pi \simeq 0.2093$, even though the auxiliary band has Chern number zero. This zero is required by comparing a nonzero global mode with the nontrivial target bundle; it is not evidence that onsite measurements prepare a Chern insulator. It illustrates why survival of overlap winding and stability of the auxiliary occupied band are different statements.

VI. FROM REFERENCE STATES TO CONDITIONED DYNAMICS

The auxiliary Hamiltonian gives a natural reference state, but it is not automatically the generator of the circuit. We can make the connection and its limitation explicit. For normalized truncated modes define the penalties

$$\hat{q}_{j,-} = \hat{\mathbf{1}} - \hat{N}_{j,-}, \quad \hat{q}_{j,+} = \hat{N}_{j,+}, \quad \hat{H}_{\text{pen}} = \sum_{j,n=\pm} \hat{q}_{j,n}. \quad (32)$$

Apart from a many-body constant, its one-body matrix is exactly $S_{+,w} - S_{-,w} = h_w$. Because each $\hat{q}$ is a projector,

$$e^{-\lambda\hat{q}} = \hat{\mathbf{1}} + (e^{-\lambda} - 1)\hat{q}, \quad e^{-\lambda\hat{q}_{j,-}} \xrightarrow{\lambda \rightarrow \infty} \hat{N}_{j,-}, \quad e^{-\lambda\hat{q}_{j,+}} \xrightarrow{\lambda \rightarrow \infty} \hat{\mathbf{1}} - \hat{N}_{j,+}. \quad (33)$$

These are the desired-outcome branches without correction. At fixed finite volume and β , weak filters obey the Trotter limit

$$e^{-\beta\hat{H}_{\text{pen}}} = \lim_{M \rightarrow \infty} \left(\prod_{j,n} e^{-\beta\hat{q}_{j,n}/M} \right)^M. \quad (34)$$

Within the half-filled sector, normalized imaginary-time evolution selects the ground state of h_w if the initial state has nonzero overlap with it and that ground state is unique. This conclusion does not apply to an initial state in the wrong number sector.

Taking each filter to be a strong projector at fixed step size is a different limit. The penalties do not commute, so Eq. (34) does not identify the strong-filter circuit with imaginary time. Further, actual records include undesired outcomes followed by correction. In the ideal fresh-ancilla protocol, filling and depletion introduce physical-layer creation and annihilation branches. An earlier corrected mode can then be disturbed by a later, nonorthogonal one. At finite range there is generally no common state that fills every lower-family mode and empties every upper-family mode.

This distinction also separates conditional dynamics from dissipative preparation of a unique pure state. Goldstein's no-go theorem and constructions concern Gaussian Lindblad preparation under specified locality and convergence conditions [7]. They do not identify our auxiliary ground state with every conditioned output. Likewise, the untruncated Markovian analysis in BPJ Appendix D is not a proof of stability of all finite-range records [1].

A. Particle addition and removal

Charge fluctuations do not invalidate the question of a bulk invariant, but they prevent a direct equal-rank comparison. To see this, take a pure Slater state with projector Γ and apply $\hat{c}^\dagger(v) = \sum_i v_i \hat{c}_i^\dagger$. Writing the state as an exterior product of occupied orbitals shows that the component Γv gives zero by Pauli exclusion. For a nonzero-probability event the surviving orbital and new projector are

$$\phi_+ = \frac{(\mathbf{1} - \Gamma)v}{\sqrt{v^\dagger(\mathbf{1} - \Gamma)v}}, \quad \Gamma' = \Gamma + \phi_+ \phi_+^\dagger. \quad (35)$$

Similarly, choose an occupied basis whose first vector is proportional to Γv . Annihilation removes precisely that vector, yielding

$$\phi_- = \frac{\Gamma v}{\sqrt{v^\dagger \Gamma v}}, \quad \Gamma' = \Gamma - \phi_- \phi_-^\dagger. \quad (36)$$

Both changes have operator norm one and change $N_F = \text{tr} \Gamma$ by one. A small *relative* number change therefore does not imply that the occupied projector is a small operator-norm perturbation.

If v is supported near one point and Γ_{ij} decays rapidly in the bulk, the numerators in Eqs. (35)–(36) inherit that locality. For a fixed allowed event the denominator is nonzero. A finite number of such localized changes is compatible with an unchanged thermodynamic bulk index: the associated change of the Fredholm operator is compact, so its index is stable when the locality assumptions defining that index remain satisfied [8]. A finite-region Chern estimator can still change near the modified orbitals.

This reasoning is not uniform over arbitrary long records. Small outcome probabilities and repeated updates can invalidate uniform locality bounds; an increasing density of modifications is not a single fixed finite-rank perturbation in the thermodynamic limit. Net charge also does not count compensating particle–hole changes. For a reference projector R , the occupation mismatches obey

$$N_p = \text{tr}[(\mathbf{1} - R)\Gamma], \quad N_h = \text{tr}[R(\mathbf{1} - \Gamma)], \quad N_p - N_h = \text{tr} \Gamma - \text{tr} R, \quad N_p + N_h = \text{tr}(\Gamma - R)^2. \quad (37)$$

The sum can be appreciable even when the difference is zero. Moreover, these are occupations relative to a chosen reference, not automatically localized defects or evidence of topological failure.

B. What the present results establish

The deterministic conclusion is that finite-range truncation need not destroy the topology of the auxiliary occupied band. We have an explicit sufficient condition and numerical checks of that condition for the shortest nontrivial square window. What remains is to control the projectors actually produced by the circuit, including their spatial decay and the long-time and large-system limits. Neither a small integrated leakage nor a relaxation statement for an averaged channel supplies this missing control.

Our existing trajectory-resolved data address a complementary question. At cycle 40, the deviation $|\overline{C_G} - 1|$ decreases from approximately 2.4×10^{-3} for $L = 20$ to 1.4×10^{-5} for $L = 40$, using square hard-wall systems, 100 trajectories, $w = 1$, and disk radius $0.2L$. The estimator is evaluated within each trajectory before averaging. This supports the intended bulk phase on the tested scales, but the comparison increases both system size and disk radius. A finite disk has finite-radius corrections and is not an exactly quantized invariant [9]. These observations should therefore be presented as dynamical numerical evidence, independently of the auxiliary-band stability argument.

Appendix A: A bound directly from normalized OW errors

It is useful to retain a more general estimate that does not use the cancellation in Eq. (20). Let $\mathcal{T}_0$ have columns $W_{r,\nu,-}/\sqrt{2}$, and let $\mathcal{T}_w$ have columns $W_{r,\nu,-;w}/\sqrt{2}$, with each truncated mode normalized to one. The tight-frame identity implies $\mathcal{T}_0\mathcal{T}_0^\dagger = P_-$ and $\|\mathcal{T}_0\|_{\text{op}} = 1$. Defining $E_w = \mathcal{T}_w - \mathcal{T}_0$ and $\epsilon_w = \|E_w\|_{\text{op}}$ gives

$$\mathcal{F}_w = \mathcal{T}_w\mathcal{T}_w^\dagger = \frac{1}{2}S_{-,w}, \quad \|\mathcal{F}_w - P_-\|_{\text{op}} \leq 2\epsilon_w + \epsilon_w^2. \quad (\text{A1})$$

Here $\mathcal{F}_w$ is a quadratic frame operator; it is not F_w . If $2\epsilon_w + \epsilon_w^2 < 1/2$, the same spectral argument as in Sec. IV proves that $\Pi_w = \mathbf{1}_{(1/2,\infty)}(\mathcal{F}_w)$ has Chern number one. This is a lower-family-only reference Hamiltonian $\mathbf{1} - 2\mathcal{F}_w$, distinct from h_w . Their occupied projectors must not be assumed equal outside a regime where that equality has been checked.

For translation-invariant modes, a bound on ϵ_w can be made independent of the volume. Write $W_\nu = W_{0,\nu,-}$, with unit real-space norm, and set

$$M_1 = \max_\nu \|W_\nu\|_1, \quad a_p = \max_\nu \|(1 - g_w)W_\nu\|_p, \quad b_w = a_1 + \left[(1 - a_2^2)^{-1/2} - 1\right] M_1. \quad (\text{A2})$$

The ℓ^1 norms here sum absolute component values over cells and orbitals; ℓ^2 is the ordinary mode norm. For $a_2 < 1$, subtracting W_ν from $g_w W_\nu / \|g_w W_\nu\|_2$ bounds the column's ℓ^1 error by b_w . Fourier transformation bounds each Bloch-column error by the same number. There are two columns divided by $\sqrt{2}$, so their combined Frobenius norm, and hence operator norm, is at most b_w . Consequently,

$$\epsilon_w = \sup_{\mathbf{k}} \|E_w(\mathbf{k})\|_{\text{op}} \leq b_w, \quad \epsilon_w < \sqrt{3/2} - 1 \implies \mathfrak{C}[\Pi_w] = 1. \quad (\text{A3})$$

Exponential localization makes M_1 finite and $a_{1,2}$ tend to zero. This supplies a sufficient finite-range result without replacing an operator bound by a per-column squared-weight estimate. It is intentionally conservative and is not used to certify the range-one example.

Appendix B: Numerical and source checks

The accompanying calculation uses the existing repository Bloch-projector and Fourier helpers without invoking circuit dynamics or their plotting entry point. It verifies equal mode norms, Hermiticity, idempotency of R_w , and the quadratic frame identity independently. Increasing the Fourier grid from 1024^2 to 2048^2 changes the retained coefficients by less than 10^{-12} ; matrix identities are checked to 10^{-12} . The momentum grids contain the Γ point, so the onsite crossing is also checked directly at its analytic s_* . We use the spectral subspace of F_w above $1/2$, not a normalized column of F_w , when constructing R_w .

For the overlap zeros, real roots along $k_y = 0$ are refined in a smooth local eigenvector frame. Their winding is checked with 360 and 720 segments on loops of radius 0.01, 0.02, and 0.04, using two distinct local gauges and explicitly nonzero boundary amplitudes. Auxiliary-band and single-mode Chern numbers are also compared using the Fukui–Hatsugai–Suzuki discretization [10] on both momentum grids. For $w = 0, 1, 2$, the auxiliary values are

respectively 0, 1, 1, while the resolved nowhere-zero compact spinors give zero. These checks guard against confusing a single mode with the auxiliary band and against assigning topology from an unresolved minimum.

The data receipt records grids, normalizations, residuals, sampled extrema, and source hashes; the plotted arrays and table are regenerated from the same calculation. The brief trajectory comparison in Sec. VI is taken from the completed slab-topology study rather than rerun here. No continuum supremum, asymptotic trajectory convergence, or localization theorem is certified by these finite-grid checks.

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